



QUALITY MANAGEMENT BRANCH

STANDARD OPERATING PROCEDURES

FOR

VERIFICATION AND CERTIFICATION OF
OZONE PRIMARY AND TRANSFER STANDARDS
USING THE STANDARD REFERENCE PHOTOMETER

Standards Laboratory SOP 5720

Second Edition

MONITORING AND LABORATORY DIVISION

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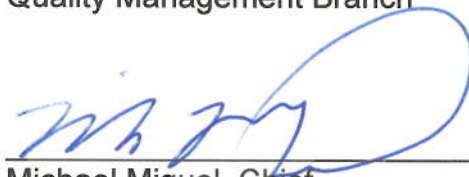
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1.0 Introduction

This standard operating procedure (SOP) describes the certification of ozone transfer standards and verification of ozone primary standards. Ozone standards are required for the calibration and auditing of ambient ozone monitors. Standard Reference Photometer (SRP) #4 was entrusted to the Air Resources Board (ARB) Standards Laboratory by the U.S. Environmental Protection Agency (U.S. EPA) to perform ozone level calibrations of level 2 and 3 ozone transfer standards. SRP #4 was manufactured by the National Institute of Standards and Technology (NIST).

2.0 Regulatory Requirements

40 CFR Appendix D to Part 50 - Measurement Principle and Calibration
Procedure for the Measurement of Ozone in the Atmosphere

- This regulation describes the calibration procedure of reference methods for measuring ozone in the atmosphere.

U.S. EPA, Quality Assurance Handbook for Air Pollution Measurement Systems, Volume II, Ambient Air Quality Monitoring Program (EPA-454/B-08-003)

- The handbook describes the Ambient Air Quality Surveillance Program and the data collection activities inherent to that program.

U.S. EPA, Transfer Standards, For the Calibration of Ambient Air Monitoring Analyzers for Ozone, Technical Assistance Document (EPA-454/B-10-001)

- This regulation defines, specifies, and formalizes the verification of ozone transfer standards for calibrating ambient ozone analyzers.

3.0 Summary of Method

The primary purpose of an ozone transfer standard is to transfer the accuracy of the ozone concentration standard of higher authority from one place and time to another. Because a transfer standard has initially no authority of its own, its authority must first be established. The essence of establishing the authority of the transfer standard is to establish a high probability or confidence that the ozone standard has reproducibility by certifying its accuracy to a higher level ultraviolet (UV) standard and then by periodically re-certifying its accuracy and stability. Certification requires the comparison between the transfer standard and SRP. Each comparison must cover the instruments full concentration range and is to be carried out on different days. Each comparison must consist of 6 or more individual comparison points, including 0 and 90 percent (± 5 percent of the upper limit of the transfer standard). The other points are to be evenly spaced between these points. This method of assaying ozone is based upon the tendency for it to absorb light at a specific wavelength (254 nanometers). The amount of light absorbed is directly proportional to the amount of ozone present. The SRP introduces a clean gas (reference gas) into the photometer and measures the amount of light (254 nanometers) through it. Then, a sample gas is introduced and the amount of light passed is measured. Based upon the above variables, an ozone concentration is then calculated. The SRP Station as used by the Standards Laboratory is represented in Figures 1 and 2.

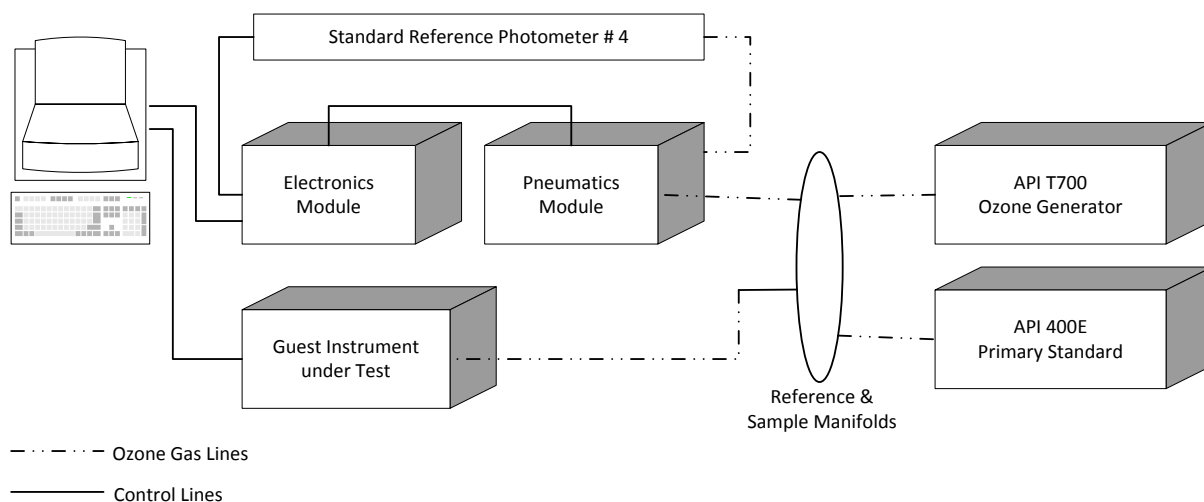


Figure 1 Schematic Diagram of SRP Station

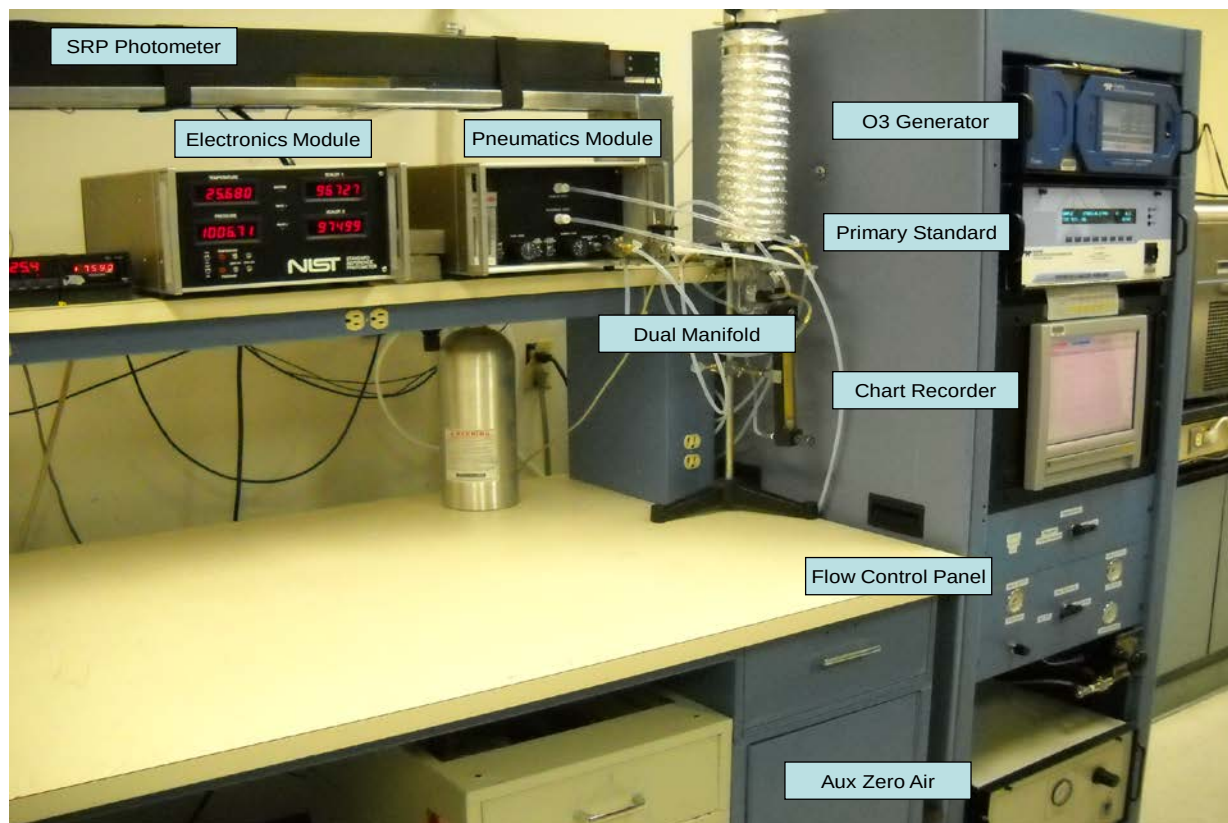


Figure 2 Photograph of SRP Station

4.0 Summary of Changes to Previous Version

- 4.1 The SOP was revised to comply with U.S. EPA's Guidance for Preparing Standard Operating Procedures (SOPs), U.S. EPA QA/G-6.
- 4.2 Incorporation of the internal SRP ozone generator as the primary ozone source and only using the external ozone generator, Teledyne Advanced Pollution Instrumentation (TAPI) T700, as an alternate. SRP software is also used to collect guest analyzer data responses.

5.0 Definitions

- Calibration – Establishes a correction formula to be used to adjust or correct the display of the candidate instrument. This is determined through a comparison of the candidate instrument to a known, or reference standard.
- Certification – Establishes traceability of a transfer standard to the SRP. The certification of an instrument requires the results of six calibrations to meet requirements established by U.S. EPA.
- Level 1 (Standard Reference Photometers) – The International Bureau of Weights and Balances maintains SRP #27 as the World's ozone reference standard. SPR #4, operated by the ARB's Standards Laboratory, is traceable to SRP #27.
- Level 2 (Primary Standards) – A device capable of assaying ozone concentrations by UV photometry in accordance with the Code of Federal Regulations, part 50, Appendix D (40CFR50). Essentially, these types of instruments are ozone photometers that require a source of reference gas separate from sample gas (including purified sample gas). Agencies are encouraged to compare primary standards to SRPs as part of their routine quality assurance programs.
- Level 3 (Transfer Standards) – A transportable device or apparatus that is capable of accurately producing or assaying ozone concentrations.
- Recertification – Establishes traceability of a transfer standard to the SRP using the current, one-time calibration along with previously completed calibrations. Recertification requirements are the same as Certification.
- Verification – Establishes a comparability of a candidate instrument to a known, or reference standard. The output of the candidate instrument is not corrected based upon the results of the verification procedures.

6.0 Personnel Qualifications

A thorough understanding of the ozone calibration process and the ability to research equipment manuals and schematics are essential. Before new personnel perform this procedure, one or more weeks of training from Standards Laboratory staff is required. They must also demonstrate competency in performing this procedure without assistance. Training from U.S. EPA personnel is highly recommended.

7.0 Health and Safety

Use proper lifting techniques or request assistance when moving the guest instruments to the SRP workbench or removing analyzers from the SRP equipment rack.

8.0 Cautions

- 8.1 It may be necessary to work around live voltage circuits when making pneumatic connections on the interior of a guest instrument.
- 8.2 Guest instruments may have different types of sample/reference inlet connectors. Ensure only like material tubing connectors are used, i.e. Teflon to Teflon or metal to metal.

9.0 Interferences

- 9.1 An isolation power transformer must be used to connect the SRP Power Module to commercial alternating current (AC) power.
- 9.2 Avoid contamination of SRP and guest instruments. Only clean, filtered gas should pass through the sample or reference manifolds.
- 9.3 Avoid over pressurization of guest instruments by ensuring that the sample and reference manifolds are vented to atmosphere. The guest instruments should be connected directly to the manifolds.

10.0 Instrument Siting Requirements

A laboratory environment with stable temperature control is acceptable. High relative humidity (e.g. > 70%) in the laboratory may impact the performance of the Stability Monitor test performed in Section 15. Relative humidity levels are monitored using the Standards Laboratory's Vaisala Transmitter, PTU301.

11.0 Equipment and Supplies

- SRP and support equipment.
- SRP isolation power transformer.
- Computer with printer.
- NIST Standard Reference Photometer, SRP Control Program for Windows NT/2000/XP.
- TAPI T700 calibrator, ozone source.
- TAPI 400E, ozone primary standard.
- Eurotherm, electronic strip chart recorder
- Ultra-pure grade air generator capable of supplying clean air at 35 pounds per square inch (psi) and 15 liters per minute (LPM).
- Teflon tubing.
- NIST traceable barometer, for room pressure reading.
- NIST traceable thermometer, for room temperature reading.

12.0 Initial SRP Setup Requirements:

- 12.1 Using the most current SRP Program from NIST, setup for the operation of the primary SRP is through the "**Edit/Setup SRPControl**" menu option as shown in Figure 3. This option brings up the setup panel where you may input the various selections that control how this program functions.

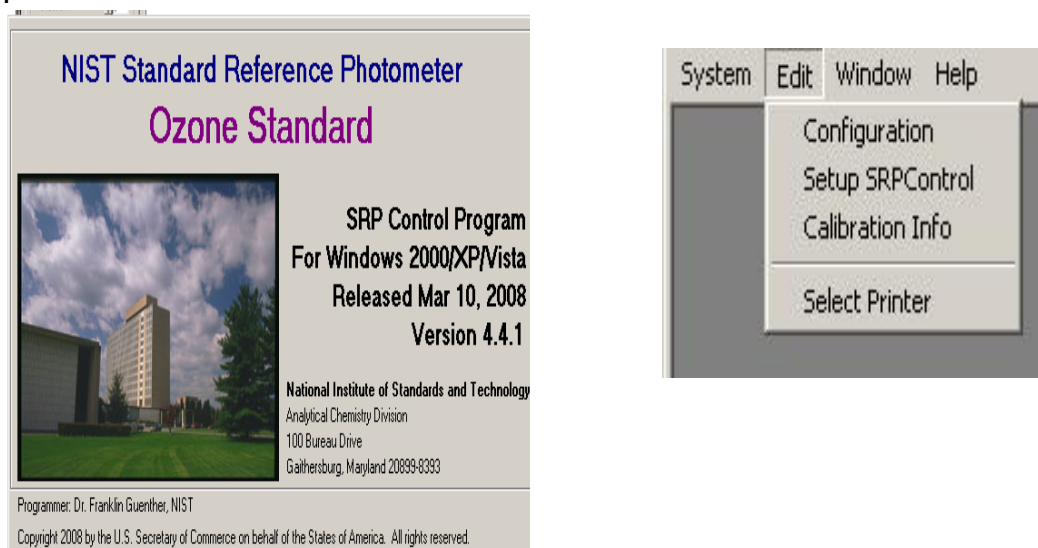


Figure 3 Edit/Setup Window

12.2 The specific SRP settings for SRP #4 are shown in Figure 4, along with the description of each setup item.

Figure 4 Setup Window

- **Digital I/O Board #:** The number assigned in *InstaCal* for the digital I/O board, usually the PCI-PIO1001/2.
- **Port Cluster #:** The port cluster (1 through 4) used for this SRP. This is set to 1 when using the PCI-PIO1001/2.
- **Analog Input Board #:** The number assigned in *InstaCal* for the analog input board, usually the PCI-PIO1001/2.
- **Analog Output Board #:** The number assigned in *InstaCal* for the analog

output board, usually the PCI-PIO1001/2.

- **SRP Serial Number:** The serial number of the main SRP, this is found on the brass panel on the photometer.
- **Cell Length (cm):** Shows the cell length for this SRP. This is not user editable.
- **Instrument Stability Factor:** The maximum standard deviation (in ppb) of four readings that will indicate that the ozone concentration is stable. Data acquisition will not start until this factor is met.
- **Data Quality Factor:** The maximum standard deviation (in ppb) allowable during actual data acquisition. Data acquisition will restart if this factor is not met.
- **Zero Flush Time (Sec):** The time in seconds that the program will wait after the ozone generator is set to zero.
- **Temp Probe Correction:** Option not used. Set to 0.0.
- **ID Label:** The label that will be used to identify this instrument in reports and in the control program.
- **External Temperature Sensors:** Option not used.

Other Settings:

- **Data Directory:** The path of the directory where the Excel reports will be saved.
- **Save Setup when exiting Program.**

13.0. SRP Setup Procedures

- 13.1 The sample lines should be initially conditioned with ozone. New sample lines may initially absorb ozone until properly conditioned and may bias the guest instrument's response.
- 13.2 Allow guest instrument to thoroughly warm-up.
- 13.3 At least once a week or following SRP maintenance, perform the Routine Service Check as outlined in Section 19.0.
- 13.4 If the API T700 is used as the ozone generation source, ensure the flow rate is greater than the sample flow rate of the guest instruments and SRP (which is approximately 5 liters per minute (lpm)). The flow setting on the API T700 is normally set to 7 lpm.
- 13.5 Enter the guest instrument information into the **Ozone Log** as shown in Figure 5. The log is located at: "W:\Standards Laboratory\Equipment Logs\Logs.mdb"

Ozone Log											
Log No	Seq	Date	Property I	Manufact	Model	Action	Agency	Type	Mo	Init	Remarks
2012-350	- 1*	12/28/2012	20004401	API	400	Re-Cert	Quality Assurance	Analyze	N	JF	
2012-361	- 1*	12/28/2012	20021064	API	401	Re-Cert	Quality Assurance	T-Std	N	JF	Van E
2012-361	- 2*	1/4/2013	20021064	API	401	Re-Cert	Quality Assurance	T-Std	N	JF	Van E
2012-361	- 3*	1/8/2013	20021064	API	401	Re-Cert	Quality Assurance	T-Std	N	JF	Van E
2013-001	- 1*	1/8/2013	20092613	API	400E	Verify	Standards Lab	P-Std	S	JF	1st Qrt Cert.
2012-361	- 4*	1/9/2013	20021064	API	401	Re-Cert	Quality Assurance	T-Std	N	JF	Van E
2013-017	- 1*	1/9/2013	107601	API	401	Re-Cert	Yolo-Solano AQMD	T-Std	N	JF	Password: 717
2012-361	- 5*	1/10/2013	20021064	API	401	Re-Cert	Quality Assurance	T-Std	N	JF	Van E
2012-361	- 6*	1/11/2013	20021064	API	401	Re-Cert	Quality Assurance	T-Std	N	JF	Van E
2013-030	- 1*	1/11/2013	107559	API	400E	Verify	Maricopa County EPA	P-Std	S	JF	
2013-030	- 2*	1/15/2013	107559	API	400E	Verify	Maricopa County EPA	P-Std	S	JF	
2013-030	- 3*	1/16/2013	107559	API	400E	Verify	Maricopa County EPA	P-Std	S	JF	
2013-030	- 4*	1/17/2013	107559	API	400E	Verify	Maricopa County EPA	P-Std	S	JF	
2013-031	- 1*	1/17/2013	20024789	API	400A	Re-Cert	Quality Assurance	Analyze	S	JF	Van X
2013-030	- 5*	1/18/2013	107559	API	400E	Verify	Maricopa County EPA	P-Std	S	JF	
2013-031	- 2*	1/18/2013	20024789	API	400A	Re-Cert	Quality Assurance	Analyze	S	JF	Van X
2013-030	- 6*	1/22/2013	107559	API	400E	Verify	Maricopa County EPA	P-Std	S	JF	
2013-038	- 1*	1/22/2013	20062228	Tanabyte	300t	Re-Cert	Special Purpose Moni	T-Std	N	JF	s/n: 0192

Figure 5 Ozone Log Entries

14.0 Guest Instrumentation Setup Procedures:

A binder labeled “Equipment Information” (the binder label is shown in figure 6) contains setup procedures for most commonly used ozone analyzers tested by the Standards Laboratory. For instruments not covered in the folder, refer to the instrument’s operation manual to obtain the following data: sample frequency, control frequency, sample temperature, sample pressure, and sample flow setting.



Figure 6 Ozone Equipment Reference Binder

15.0 SRP Diagnostic/Calibration Procedures:

15.1 Diagnostic scripts can be accessed under the "**System/Diagnostic**" menu option as shown in Figure 7.

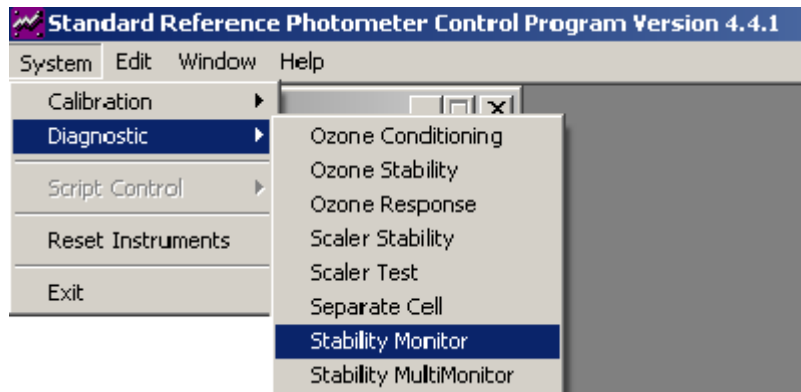


Figure 7 System/Diagnostic Window

15.2 The available scripts are:

- Ozone Conditioning: This diagnostic can test all instruments. It lists the ozone concentration from all instruments at a set lamp percent. It runs until you click stop or abort.
- Ozone Stability: This diagnostic test runs only on SRPs. It lists the scaler1, scaler2, and Ozone Concentration. You may run up to 100 repeat readings in up to 100 cycles.
- Ozone Response: This diagnostic test runs only on SRPs. It calculates the ozone concentration in each cell, and the ratio of those concentrations. The lamp starts at zero, ramps to full, then back to zero.
- Scaler Stability: This diagnostic test runs only on SRPs. It lists the scaler1, scaler2, and scaler ratio. You may run up to 100 repeat readings in up to 100 cycles.
- Scaler Test: This diagnostic test runs only on SRPs. It lists the temperature, pressure, scaler1, scaler2, and scaler ratio. It runs until you click the stop or abort button.
- Separate Cell: This diagnostic test runs only on SRPs. It calculates individual cell ozone concentrations, and the ratio of those concentrations. You may run up to 100 repeat readings in up to 100 cycles.
- Stability Monitor: This diagnostic test runs only on SRPs. It lists the Temperature, Pressure, scaler1, scaler2, and scaler ratio. You may run up to 100 repeat readings in up to 100 cycles. This diagnostic generates an Excel report using the DiagReport.xls template.
- Stability MultiMonitor: This function is only used during the annual SRP to SRP verification. Specific instruction is included with the verification SRP.

- 15.3 To start the Stability monitor, click **SRP 4** and **Print Status Report Window** as shown in Figure 8. After 20 complete cycles, the desktop will look like the illustration as shown in Figure 9. This routine is performed daily.

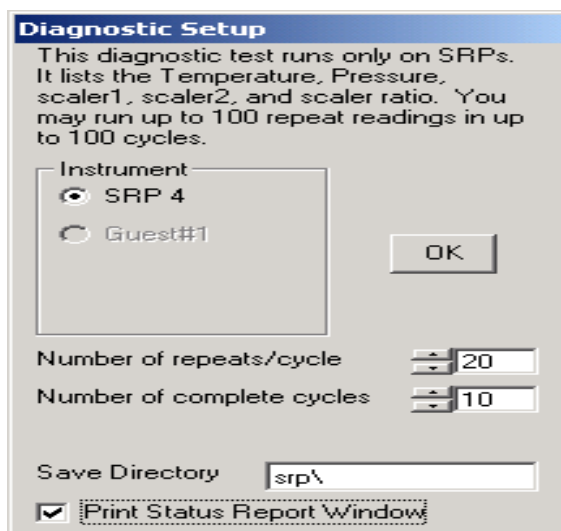


Figure 8 Diagnostic Setup Window

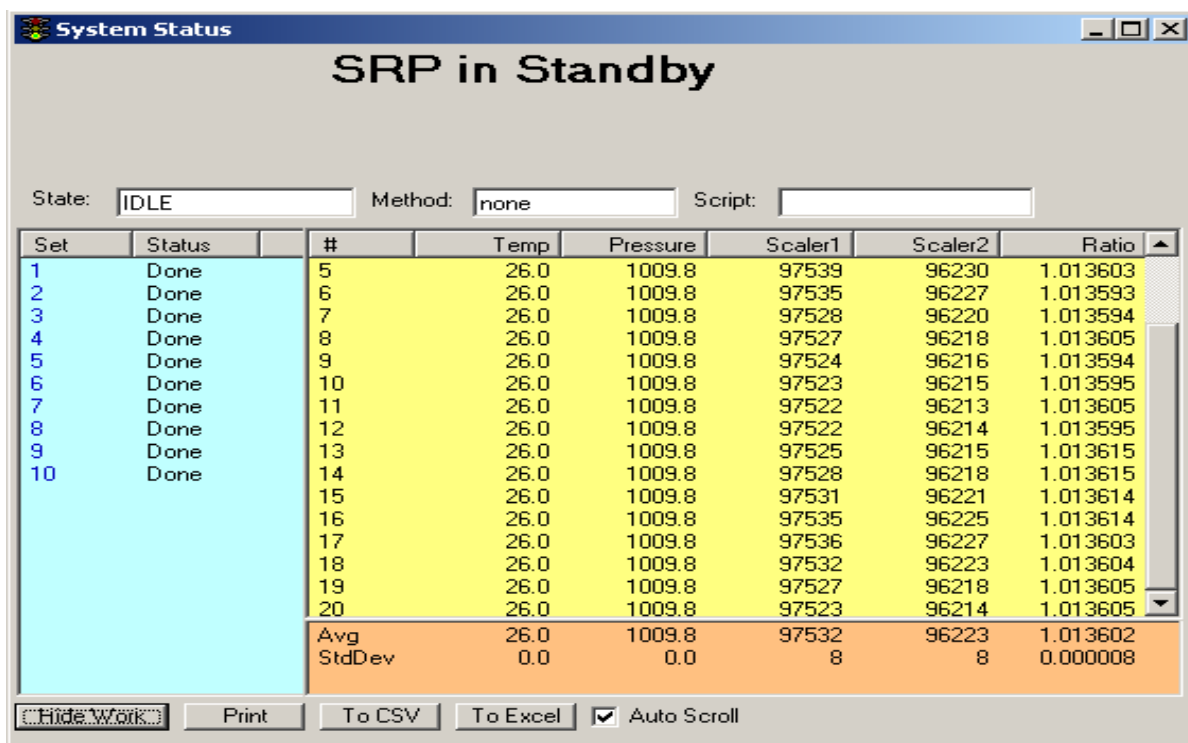


Figure 9 Diagnostic Results

- 15.4 At the conclusion of the Stability monitor test, the printer will output a hard copy of the Diagnostic Test Report as shown in Figure 10.

- 15.5 The performance criteria of the test are determined from the last three lines (8 through 10). The **Std Dev** between scaler 1 and scaler 2 must be less than 25. The **Std Dev Ratio** between scaler 1 and scaler 2 must be less than 0.000030. If the diagnostics fails to meet these standards, perform the test again or determine the reason for failure.
- 15.6 From line 10, record the temperature, pressure and scaler 1 and scaler 2 values into the "SRP #4" section on the form "Ozone Data Entry Form." The data sheet is illustrated in Section 16.17, Figure 22.

Diagnostic Test Report					
Stability Monitor					
Calibrating Institute:	CARB			Date:	7/21/2011
Operator:	Jerry Freeman			File Name:	dstab128.xls
Instruments:	SRP #4				
Comment:	Reps = 20; Cycles = 10				
	Temp	Pressure	Scaler1	Scaler2	Ratio
1 Average:	26.11	1009.9	97596	96296	1.013500
Std Dev:	0.00	0.0	7	7	0.000011
2 Average:	26.11	1009.9	97619	96321	1.013479
Std Dev:	0.00	0.0	8	8	0.000008
3 Average:	26.09	1009.8	97638	96342	1.013449
Std Dev:	0.01	0.0	9	9	0.000009
4 Average:	26.07	1009.8	97643	96347	1.013451
Std Dev:	0.01	0.0	9	9	0.000007
5 Average:	26.05	1009.8	97640	96341	1.013485
Std Dev:	0.01	0.0	12	12	0.000012
6 Average:	26.03	1009.8	97624	96323	1.013509
Std Dev:	0.01	0.0	6	7	0.000012
7 Average:	26.02	1009.8	97601	96298	1.013530
Std Dev:	0.01	0.0	8	8	0.000008
8 Average:	26.01	1009.8	97586	96281	1.013550
Std Dev:	0.00	0.0	13	13	0.000007
9 Average:	26.01	1009.8	97565	96259	1.013572
Std Dev:	0.00	0.0	12	13	0.000013
10 Average:	26.02	1009.8	97532	96223	1.013602
Std Dev:	0.01	0.0	8	8	0.000008

Figure 10 Diagnostic Test Report

16.0 Ozone Assay Test Operation:

- 16.1 Ensure the guest instrument has sufficiently warmed-up and has stabilized.
- 16.2 Connect a sample line labeled "Ozone" to the sample inlet port on the back of the guest instrument.
- 16.3 Consult the equipment binder to determine the reference air input requirements. As a rule, a primary standard instrument requires clean air at room pressure from the gas line labeled "Low Zero Air". Transfer standards require pressurized clean air supplied by the gas lines labeled "High Zero Air".

- 16.4 Obtain room pressure/temperature readings from a NIST traceable source. The Visalia Pressure/Temperature Translator is one of the instruments available in the Standards Laboratory. It is located near the Humidity Station.
- 16.5 Perform instrument set-up procedures; see the equipment binder for specific instrument model.
- 16.6 On the Flow Control Panel (see Figure 2), adjust the air pressures as indicated:
- House zero air system to 35 – 40 psi.
 - TAPI T700 input to 25 – 35 psi.
 - SRP input to 25 psi.
- 16.7 To configure the instrument under test, click on **Edit** then **Configuration** to reveal the Instrument Configuration screen as shown in Figure 11.

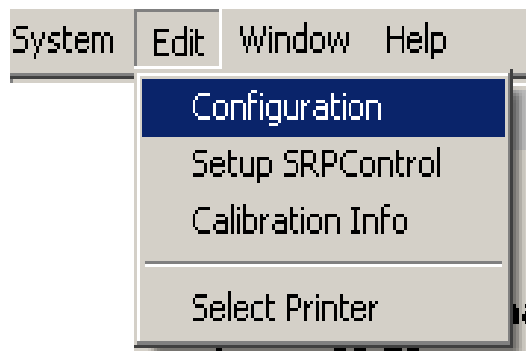


Figure 11 Edit/Configuration Window

- 16.8 Enter the following typical information into the **Instrument Configuration** panels as shown in Figures 12 through 14:

NOTE: For detailed information concerning these entries, refer to U.S. EPA's SRP SOP, section 5, Guideline for Guest Instrument Configuration. This SOP is listed in Section 22, References. Not all fields are appropriate for all processes of this SOP.

- **Guest #1, Guest #2, Guest #3:** Configuration for three instruments under test. Only two instruments may be tested simultaneously using the Standards Laboratory's setup.
- **Instrument Active:** Check this box for the guest ozone analyzer under test.
- **Make:** Of the instrument under test.
- **Model:** Of the instrument under test.
- **Serial #:** ARB barcode or Standard's Laboratory number.
- **ID Label:** Unique name or the current Log Number.
- **Owner:** Of the instrument under test.
- **Contact:** Person for the instrument under test.
- **Instrument Stability Factor:**
 - Acceptable range: 0.7 (normal) to 2.0 (extremely unstable).
- **Data Quality Factor:**
 - Acceptable range: 0.7 (normal) to 2.0 (extremely unstable).
- **Duty Cycle (Secs):** The time in seconds the instruments takes to update the ozone reading (normally 5 to 10 seconds).
- **Connection Mode:** Three modes are available for instrumentation testing. Panel illustrations are shown in the indicated figures, followed by directions for "Configuration Connections".
 - **Serial Communications Port,** Figure 12.
 - **Analog Input Port,** Figure 13.
 - **SRP Digital I/O,** Used only during SRP verification, generally against SRP #7. Detailed instructions are included in the shipping container.
 - **None,** Figure 14.
- **Activate:** To load the Instrument Configuration data.

Instrument Configuration

Guest #1 Guest #2 Guest #3

Configure Instrument #1

☒ Instrument Active

Make: API

Model: 400E

Serial #: 20092613

ID Label: API 400E D

Owner: CARB

Contact: Jerry Freeman

Instrument Stability Factor: 1.0

Data Quality Factor: 1.0

Duty Cycle (Secs): 6

Connection Mode

☒ Serial Communications Port

☐ Analog Input Port

☐ SRP Digital I/O

☐ None

Configure Connection

Comm Port#: 1

Baud Rate: 9600

Bits, Parity, Stop: 8, n, 1

Inst Driver File: API400A.drv

Instrument ID: (ASCII code = \#)

Password: 940331 O3 Generator: ☐

Activate Save Load

Figure 12 Instrument Configuration – Serial Mode

- **Comm Port #:** Select either Comm Cable 1 or 2. The cables are located at the left side of the work bench. The instruments must be configured to match the SRP. Detailed instructions are located in the “Equipment Information” binder for certain instrument models. Not all models are serial capable.
- **Baud Rate:** The setting is always: 9600.
- **Bits, Parity, Stop:** The setting is always: 8, n, 1.
- **Inst Driver File:** From the pull-down menu, select the driver that matches the instrument's make and model. Drivers are not available for all models.
- **Instrument ID (ASCII code = \#):** \187. This code is used for Thermo Electronic, Model 49i series instruments only. For all others, leave blank.
- **Password:** 940331. This code is used for TAPI instruments only. For all others, leave blank.
- **O3 Generator:** This option is not used. Leave unchecked.

The image shows a software window titled "Instrument Configuration" with three tabs: "Guest #1", "Guest #2", and "Guest #3". The "Guest #2" tab is selected. Inside the tab, there is a section titled "Configure Instrument #2". On the left side of this section, there is a checkbox labeled "Instrument Active" which is unchecked. Below it are several text input fields: "Make" (Sabio), "Model" (2010D), "Serial #" (11800513), "ID Label" (2013 214), "Owner" (AM - North), and "Contact" (empty). Below these fields are three numeric input fields with up/down arrows: "Instrument Stability Factor" (1.0), "Data Quality Factor" (1.0), and "Duty Cycle (Secs)" (5). At the bottom of this section are three buttons: "Activate", "Save", and "Load". On the right side of the "Configure Instrument #2" section, there is a "Connection Mode" group box containing four radio buttons: "Serial Communications Port", "Analog Input Port" (which is selected), "SRP Digital I/O", and "None". Below this is another group box titled "Configure Connection" containing three numeric input fields with up/down arrows: "Analog Channel#" (1), "Volts" (1), "Conversion Slope" (1), and "Conversion Intercept" (0).

Figure 13 Instrument Configuration – Analog Mode

- **Analog Channel #:** Select Channel 1
- **Volts:** Set to closely match instrument's output voltage.
- **Conversion Slope:** Determined in an Excel Spreadsheet named "Analog Slope-Intercept" by running three points plus zero. The spreadsheet is located at: C:\Documents & Settings\Administrator\My Documents\SRP\03-Ozone Assay Data Sheets.
- **Conversion Intercept:** Determined in an Excel Spreadsheet named "Analog Slope-Intercept" by running three points plus zero.

Instrument Configuration

Guest #1 **Guest #2** Guest #3

Configure Instrument #2

☒ Instrument Active

Make: Dasibi
Model: 1008-PC
Serial #: 107241
ID Label: Dasibi
Owner: Desert Research
Contact:

Connection Mode

☐ Serial Communications Port
☐ Analog Input Port
☐ SRP Digital I/O
☒ None

Configure Connection

Instruments with no communications connection can be calibrated by designating "none" as the Connection Mode. Data is then entered manually into the relevant Excel Spreadsheet.

☒ Data entered during acquire

Activate Save Load

Figure 14 Instrument Configuration – None

- **Data entered during acquire:** Ensure this option is checked.

16.9 Once the information is entered in the **Instrument Configuration** panel, click on the **Activate** button.

16.10 Click on **System**, then **Calibration**, then **Standard**, as shown in Figure 15.

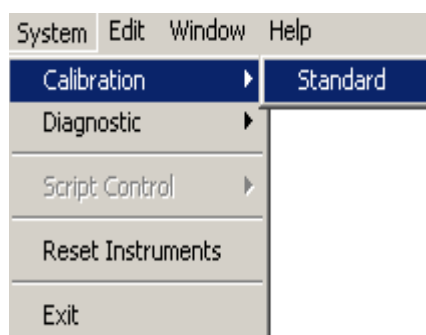


Figure 15 Initialize Calibration

16.11 Complete the **Calibration Information and Setup** panels as shown in Figures 16 and 17. These figures represent ozone instruments configured to ranges 500 ppm and 1000 ppm, respectively. Paragraph 16.12 contains an explanation for each entry.

Calibration Information and Setup

Calibrating Institute: CARB Date: 7/21/2011
Operator: Jerry Freeman File Name: C0721001.xls
Calibration Method: SL 450 PPM Data Subdirectory:

Comments:

Air Flow Rate: 6.2
Lamp Percent Range: Low 5.0 High 38.0
Lamp Percent: Conditioning 0.0
Time (min) 0
Number of Conc Points: 8
Points/Conc 10
Number of Cycles 3

Step	Generator%
1	0.00
2	38.00
3	31.40
4	24.80
5	18.20
6	11.60
7	5.00
8	0.00

☒ Order High to Low Conc
☐ Randomize Steps
☒ Zero Point at Start and End
☒ Save Raw Data in Excel
☐ Independent Data
☒ Dark Count 10
☒ Auto Print Excel Report

Instruments:
☒ API 400E D
☐ Dasibi
☐ API 703E

Excel Report Template: SRPTemplate.xls Link Method: None

Configure Instruments Manually Edit Method File Save Conditions OK Cancel

Figure 16 Calibration at 500 PPM Range

- **Calibration Method:** From the pull-down menu, select 450.

Calibration Information and Setup

Calibrating Institute: CARB Date: 7/27/2011
Operator: Jerry Freeman File Name: C0727001.xls
Calibration Method: SL 900 PPM Data Subdirectory:

Comments:

Air Flow Rate: 6.2
Lamp Percent Range: Low 5.0 High 69.0
Lamp Percent: Conditioning 0.0
Time (min) 0
Number of Conc Points: 8
Points/Conc 10
Number of Cycles 1

Step	Generator%
1	0.00
2	69.00
3	56.20
4	43.40
5	30.60
6	17.80
7	5.00
8	0.00

☒ Order High to Low Conc
☐ Randomize Steps
☒ Zero Point at Start and End
☒ Save Raw Data in Excel
☐ Independent Data
☒ Dark Count 10
☒ Auto Print Excel Report

Instruments:
☐ API 400E D
☐ API 401 A
☐ API 703E

Excel Report Template: SRPTemplate.xls Link Method: None

Configure Instruments Manually Edit Method File Save Conditions OK Cancel

Figure 17 Calibration at 1,000 PPM Range

- **Calibration Method:** From the pull-down menu, select 900.

16.12 The main section of the form is for creation of calibration methods:

- Comments: 2 lines of comments which will print on the report
- Air Flow Rate: Flow range 5 – 7 lpm.
- Lamp Percent Range - Low:
 1. Guest Instrument range 500 ppm, enter = example 5%*; (25 ppm).
 2. Guest Instrument range 1,000 ppm, enter = example 8%* (50 ppm).
- Lamp Percent Range - High:
 1. Guest Instrument range 500 ppm, enter = example 48%*; (450 ppm).
 2. Guest Instrument range 1,000 ppm, enter = example 84%* (900 ppm).
- Lamp Percent - Conditioning: Not normally used.
- Lamp Percent - Conditioning Time: Not normally used.
- Number of Conc Points: Enter 8.
- Points/Conc: Set to 10.
- Number of Cycles: Set to 1.
- Excel Report Template: SRPTemplate.xls
- Order High to Low Conc: Select.
- Randomize Steps: Not used.
- Zero Point at Start and End: Select.
- Save Raw Data to in Excel: Ensure this option is selected.
- Independent Data: Not Used.
- Dark Count: Select this option and set to 10.
- Auto Print Excel Report: Ensure this option is selected.
- Instruments: Only active instruments are available for calibrations.
Check the instruments you want to include in this calibration run
- Link Method "None".

* The condition of the ozone generator lamp will determine the Lamp Percent Range needed to reach the target values of 450 and 900 ppm. The exact lamp percentage range can be determined by using the System Diagnostic's option "Ozone Conditioning". Refer to Section 15.1.

- 16.13 On the Pneumatics Module, switch on the **Pump**, **Flow Control**, and **Ozone Generator**. Adjust the air flow in Cell 1 and Cell 2 to 20 cc/min. Refer to Figure 18.



Figure 18 Pneumatics Module

- 16.14 Under normal conditions, ozone is supplied from the internal ozone generator in the Pneumatics Module. Position the Ozone Source Selector Valve towards the **Ozone Sample Out** position as shown in Figure 19. In cases where the Alternate Ozone Generator is used, switch the valve position to the TAPI T700 position and follow the procedures outlined in Section 16.15.



Figure 19 Ozone Source Selector Valve

16.15 Alternate Ozone Generator:

- The primary source of ozone is supplied by the SRP's internal ozone generator which is program controlled.
- The TAPI T700 may be used as an alternate source for ozone generation. The Connection Mode must be selected as "**None**" in the SRP Program.
- Setup the TAPI T700 according to the following directions:
 - o Switch the Ozone Source Selector Value, located on the front of the SRP Pneumatic Module from **Ozone Sample Out** to the TAPI 700 position.
 - o On the T700's touch screen, press **Gen**, then **Auto**. Enter the appropriate ozone concentration (matching the guest instrument's range), in ppm, using Figure 20 as a guide, and then press **ENTR**. Set the total flow at 7.0 lpm, and then press **ENTR**. Repeat this process until the SRP Program is complete.

	Air Flow Less than 7 LPM						
O ₃ Percentage		90%	75%	55%	40%	20%	5%
HIGH - O ₃ Target ppm	0	900	750	550	400	200	50
O ₃ GEN Drive	0	1991	1743	1440	1236	981	741

LOW - O ₃ Target ppm	0	450	375	275	200	100	27
O ₃ GEN Drive	0	1295	1208	1088	981	844	635

Figure 20 Ozone Test Points

- 16.16 Once the information is entered into the **Calibration Information and Setup** panel, click on the **OK** button as shown in Figures 16 and 17. The SRP program automatically begins its **Preliminary Dark Count** routine which takes approximately 10 minutes. Once this routine is completed, the program will automatically begin if either the Connection Mode **Serial or Analog** were selected. If Connection Mode **None** was selected, the program will wait for manual entries of the Ozone Readings from the Guest Instrument. Figure 21 shows an impending assay for two guest instruments.

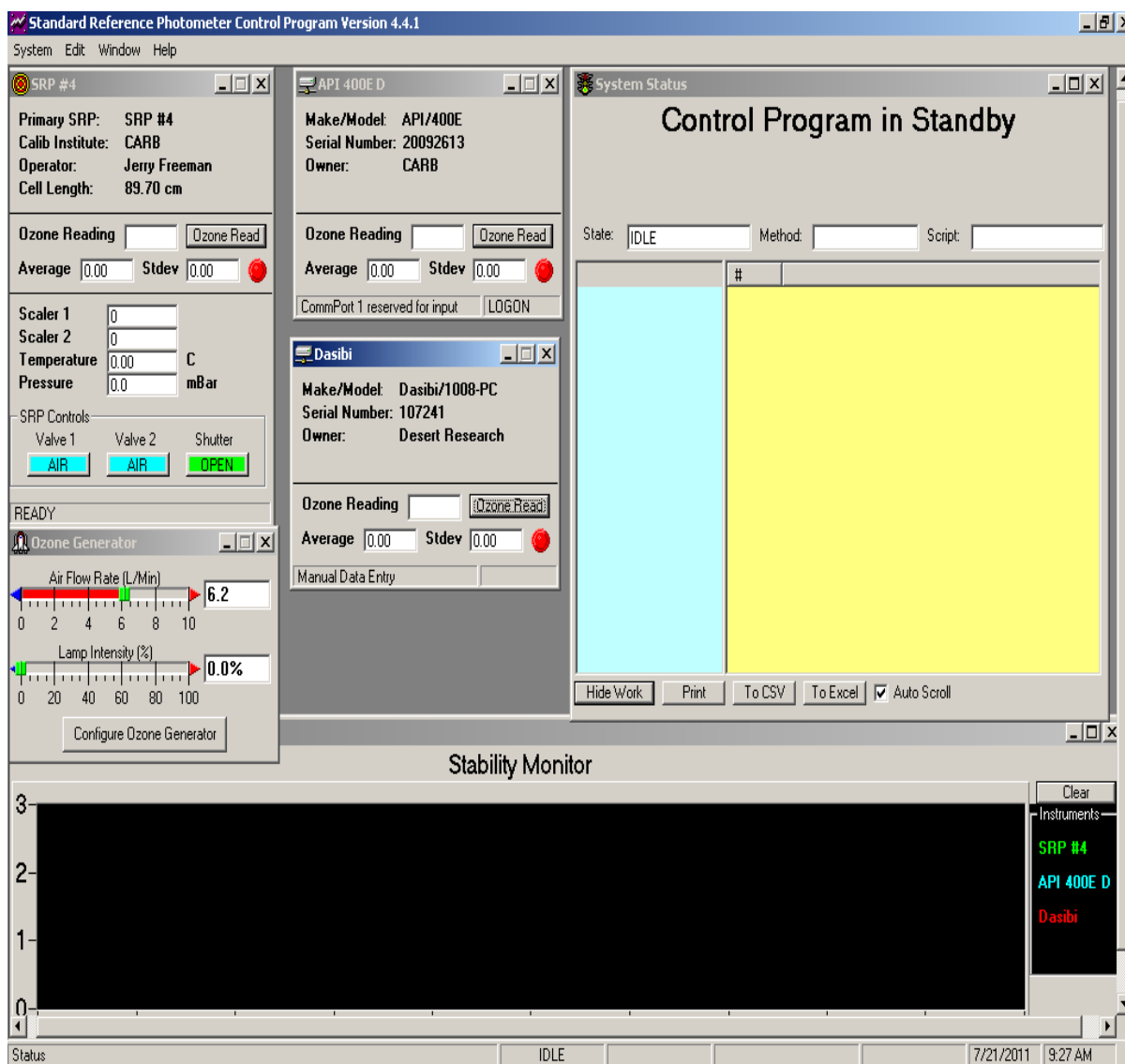


Figure 21 Calibration Display

- 16.17 The Standard Reference Photometer Control Program uses Excel to process the collected ozone data and running a second Excel program may interfere with this process. To eliminate this problem, a manual Data Entry form, Figure 22, may be used. Once the program has completed, the data can be entered in the Excel Form shown in Section 17.

Ozone Data Entry Form

LOG NUMBER: 2013 -		File #:
Instrument Model:		
Property / Serial #:		
Sample / Photometer:		
Control / Reference:		
Air Flow:		
Chamber Temp:	<u>Range:</u>	
Chamber Press:		
Offset:	Analyzer/Trans ()	
Slope:	Primary Stnd ()	

Date:
Temp:
Press (mb):
Scaler 1:
Scaler 2:
Am Press (mmHg):
Am Temp @:
<u>Analog:</u>
Slope:
Intercept:

LOG NUMBER: 2013 -		File #:
Instrument Model:		
Property / Serial #:		
Sample / Photometer:		
Control / Reference:		
Air Flow:		
Chamber Temp:	<u>Range:</u>	
Chamber Press:		
Offset:	Analyzer/Trans ()	
Slope:	Primary Stnd ()	

Date:
Temp:
Press (mb):
Scaler 1:
Scaler 2:
Am Press (mmHg):
Am Temp @:
<u>Analog:</u>
Slope:
Intercept:

LOG NUMBER: 2013 -		File #:
Instrument Model:		
Property / Serial #:		
Sample / Photometer:		
Control / Reference:		
Air Flow:		
Chamber Temp:	<u>Range:</u>	
Chamber Press:		
Offset:	Analyzer/Trans ()	
Slope:	Primary Stnd ()	

Date:
Temp:
Press (mb):
Scaler 1:
Scaler 2:
Am Press (mmHg):
Am Temp @:
<u>Analog:</u>
Slope:
Intercept:

Figure 22 Data Entry Form

- 16.16 The program is accepting data with the first point being at "0". Enter the guest instrument reading into the "Ozone Reading" window. The program allows 30 seconds to enter the data. Once the readings have stabilized, indicated by a Stdev of less than 0.78 of both the SRP reading and the instrument being assayed, the red indicators will turn green. At this point, the program will indicate **Actual Data**. After ten entries, the status column will indicate complete and the **Enter Ozone Reading** will indicate done. Repeat this process for the remaining ozone concentrations. Upon completion of the data points, the following results will be printed. An example is shown in Figure 23.

Standard Reference Photometer Calibration Report						
Calibrating Institute:		CARB			Date:	01-Mar-13
Operator:		Jerry Freeman			Start Time:	7:12
Instrument:		SRP #4 Cell Length=89.70			End Time:	8:31
Comment:					Filename:	c0301003.xls
Calibrated Instrument:		2013 072		Calibration Results	Value	Standard Uncertainty
Owner:		Clark County DAQ		Slope	1.01268	0.00031
Contact:		Mickey Palmer		Intercept	0.01841	0.07617
Make:		API		Covariance		-2.2149E-08
Model:		T400		Res Std Dev	0.13263	
Serial Number:		106018				
Calibration Parameters: Zero Start&End;Raw Saved;Dark Count On (10)						
Air Flow Rate:		6.0 l/min				
Lamp Intensity Range:		0.0 to 47.0 %				
Number Conc. Points:		8		Points/Concentration:	10	
Conditioning:		none				
Calibration	SRP #4		2013 072		2013 072	
Data Points	Result	Std. Dev	Result	Std. Dev	Predicted	Residual
Dark Count 1	9					
Dark Count 2	8					
1	0.1	0.2	0.1	0.1	0.09	0.01
2	452.8	0.2	458.5	0.2	458.53	-0.05
3	367.5	0.4	372.2	0.2	372.13	0.09
4	285.0	0.3	288.8	0.3	288.65	0.15
5	206.8	0.2	209.3	0.1	209.42	-0.13
6	125.2	0.2	126.6	0.2	126.82	-0.21
7	26.9	0.2	27.2	0.2	27.24	-0.03
8	0.0	0.2	0.2	0.1	-0.01	0.17

Figure 23 SRP Calibration Report

17.0 Ozone Data Collection Procedures:

- 17.1 If the guest instrument(s) require a re-certification, transpose the data from SRP #4 Results and Guest Instrument Result onto the Ozone Instrument Assay Data Sheet as illustrated in Figure 24. The form is located at: "C:\My Documents\SRP".

NOTE: The SRP computer is not connected to our network drives due to operational limitations and backups are performed manually.

MLD STANDARDS LABORATORY OZONE INSTRUMENT ASSAY DATA SHEET

Log Number: 2013 064
Previous Log No.: 2011-198

Date: 2/22/2013 ← Ctrl ;

Guest Ozone Instrument

Instrument Model:	API 400
Property No.:	107509
Serial No.:	537
Initial Samp/Photo:	3493.3
Final:	3492.9
Initial Cont/Ref:	3493.3
Final:	3492.9
Air Flow:	846
Final:	845
Chamber Temp C:	27.4
Chamber Press:	30.1
Offset:	-0.3
Slope:	1.026

File No.:	2013 064	-6
Analyzer/Transfer Standard:		
Primary Standard:	X	
Analyzer Range (ppb):	500	

SRP # 4

Temperature C:	22.9
Pressure:	1022.6
Scaler 1:	94955
Scaler 2:	92641
Pres. (P _A) mm Hg:	767.3
AM Temp (T _A) C:	21.8

Connection Mode - Serial:	
None:	X
Analog:	
Slope:	Intercept:

Slope:	1.001193
Intercept w/Diff.:	-0.000260
Correlation Coefficient:	0.9999981

RSD Slope	0.20	(<1.5)
FRSD Intercept	0.02	(<0.5)
Change - previous	0.09	(<1.0)

Comparison	Date	Slope	Intercept
1	2/14/2013	1.00268	-0.00004
2	2/15/2013	1.00155	0.00011
3	2/19/2013	0.99670	0.00008
4	2/20/2013	1.00081	0.00011
5	2/21/2013	1.00024	0.00018
6	2/22/2013	1.001193	-0.000260
Mean Slope		1.000527	
Mean Intercept			0.000029

Calibration Data

Air Flow	6.0 LPM						
Approx. %	0	90%	75%	55%	40%	20%	5%
Guest Data ppb	-0.1	444.8	361.7	280.3	203.4	123.9	26.4
Guest Data ppm	0.000	0.445	0.362	0.280	0.203	0.124	0.026

SRP Data ppb	0.0	444.8	361.4	280.2	203.4	123.8	26.5	
--------------	-----	-------	-------	-------	-------	-------	------	--

Difference

0	0.0
445	445.0
362	362.0
280	280.0
203	203.0
124	124.0
26	26.0

Calibrated by: Jerry

Checked by: _____

Figure 24 Ozone Data Sheet

18.0 Data Acquisition and Storage

- 18.1 Follow the displayed data path and open the “Ozone Cal Data.xlsm workbook.”
See Figure 25.

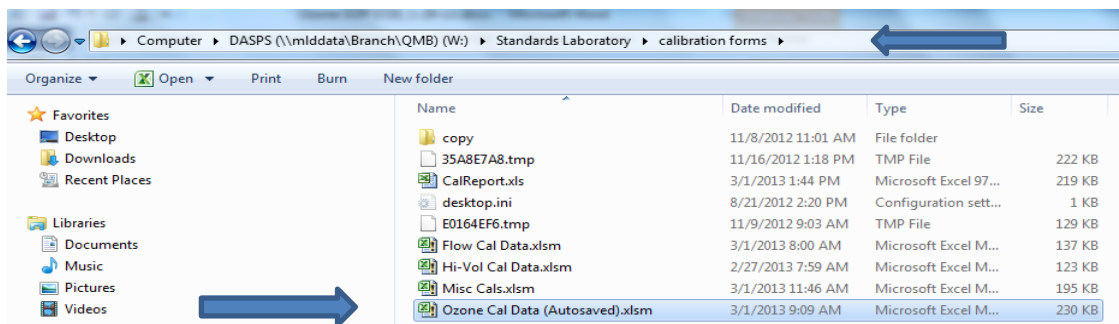


Figure 25 “Ozone Cal Data.xlsm” workbook

- 18.2 Open the “Login” worksheet. See Figure 26.

	A	B	C	D	E	F	G
1	STANDARDS LABORATORY						
2	PROJECT CHECK LIST						
3							
4	Agency:	STANDARDS LABORATORY	Customer ID				LOGIN (1)
5	Contact:	JERRY FREEMAN	Instrument ID				20092613
6	Year&Number	2013 0	Prev. Log. No.:	2012260			
7	Instrument:	API 400E, PRIMARY STANDARD					
8	Bar Code:	20092613	SN:	2271			
9	PHONE #:	916-324-1845	Date:	3/1/2013			
10	Log. Number:	2013001					
11	Select Service:	Calibration (C)	Verification (V)				
12		Enter Letter Selection		C			Collect Customer Information (3)
13	OZONE:	Calibration Date (s)					
14		1 _____ []	2 _____ []	3 _____ []			
15							
16		4 _____ []	5 _____ []	6 _____ []			
17							

Figure 26 “Login” worksheet

- 18.3 To initially login an instrument into the Standards Laboratory, enter the “Instrument ID” number into cell G4. Then press “LOGIN” as shown in Figure 26.
- 18.4 To enter calibration data, enter the “Instrument ID” number into cell G4. Then press “Collect Customer Information”. This is also shown in Figure 26.

- 18.5 Enter the calibration data collected in Section 16.1 into the appropriate blocks of the Ozone Calibration Data worksheet as shown in Figure 27. Print this report and file in the log folder.

	A	B	C	D	E	F	G	H	I	J
1	STANDARDS LABORATORY									
2										
3	OZONE CALIBRATION DATA									
4										
5	Log Number:		2013001							
6	Previous Log num:		2012260							
7	Calibration Date:		3/1/2013							
8	Property Number:		20092613							
9	Bar Code Number:		20092613							
10	Instrument Make & Model:		API 400E, PRIMARY STANDARD							
11	Property of:		STANDARDS LABORATORY							
12										
13										
14	Ozone Measurement:		Initial:	4596.0	Final:	4593.6	Zero:	-2.5		
15	Ozone Control:		Initial:	4595.9	Final:	4593.4	Span:	0.994		
16	Chamber Temperature:		39.9		°C					
17	Chamber Pressure:		29.5							
18					Air Flow Rate:		775			
19					Air Flow Sett:		774			
20	Analyzer:				Photometer:		X			
21										
22	Ozone Set Points:		0	90%	75%	55%	40%	20%	5%	
23	Guest Average (ppm):		0.000	0.459	0.372	0.289	0.209	0.127	0.027	
24	Guest Net Meter (ppb):		0.000	459	372	289	209	127	27	
25	SRP True Ozone (ppb):		0.100	452.8	367.5	285	206.8	125.2	26.9	
26	SRP True Ozone (ppm):									
27										
28										
29										
30	CALIBRATION EQUATION									
31										
32	Net Meter =		1.013621		x True Ozone		-0.0001780		ppm	
33										
34	Correction Coefficient =		0.9999985							
35										
36										
37			SLOPE	INTERCEPT	CORR. COE.					
38			PASS	PASS	PASS					
39										
40										
	O3_Cal_Data Cert Report w-gas Cert Report Ozone Cert Report V									

Figure 27 Ozone Calibration Data

18.6 Standards Laboratory Criteria for Ozone Standard Evaluations.

1. Guest instrument Full Certification service is required if the standard has been repaired, is new, or has been factory calibrated since the last time it was in for testing. Six tests must be performed with each being on different days.
2. Guest instrument Recertification service requires one run if the standard has 6 previous certifications recorded in the instrument database.

NOTE: During Recertification, if a guest instrument fails to meet criteria, a determination must be made to either repair and/or perform a Full Certification.

Requirements for Level 2 Transfer Standards (Primary Standard)-

- Slope of current test must be **within 3% of one**.
- Intercept must be within **±3 ppb**.
- Each Correlation coefficient: must be **0.9999 or better**.
- Six tests ran on different days.
- Recertification period is annually.

Requirements for Level 3 Transfer Standards-

- Relative standard deviation (RSD) slope of 6 previous slope values must be: **< 1.5** percent.
- Full scale relative deviation (FSRD) of 6 previous intercept values must be: **< 0.5** percent.
- Slope shift from previous slope must be: **< ± 1** percent.
- Slope of current test must be **within 5% of one**.
- Calibration: Intercept of current test must be **< 1% of full scale**.
- Each correlation coefficient must be 0.9999 or better.
- Calibration: Intercept must be **within ±3 ppb**.
- Recertification period is every six months.

- 18.7 Open the "CertSummary" and click the highlighted Run button. This will fill in the previous five comparisons from the database. A completed Certification Summary Sheet is shown in Figure 28. Print this report and file in the log folder.

STANDARDS LABORATORY

CERTIFICATION SUMMARY SHEET

Run 

Date: 7/15/2013
Log Number: 2013191
Property Number: 20092613
Bar Code Number: 20092613
Instrument Make & Model: API 400E, PRIMARY STANDARD
Property of: STANDARDS LABORATORY

Full Certification: ☐ Re-Certification: ☒

Comparison	Date	Slope	Intercept
1	7/3/2013	0.999221	0.00033
2	7/9/2013	1.002179	0.00020
3	7/10/2013	1.002522	0.00049
4	7/11/2013	1.001389	0.00037
5	7/12/2013	1.002452	0.00027
6	7/15/2013	1.003093	0.00019

MEAN Slope: 1.001809
MEAN Intercept: 0.000309

RSD Slope [(Stdev / Mean) x 100] = 0.14% (< 1.5)
FRSD Intercept [(Stdev / Fullscale) x 100] = 0.01% (< 0.5)
Change From Previous Calibration = 0.06% (< 1.0)

NEW CERTIFICATION RELATIONSHIP

Correlation Coef. 0.9999986

Net Display = 1.001809 x True Ozone 0.000309

True Ozone = 0.99819 x Net Display - 0.000308

	SLOPE	INTERCEPT	CORR. COE.
Calibration	PASS	PASS	PASS
Verification	PASS	PASS	PASS

 **CertSummary** **LOGIN**

Figure 28 Certification Summary Sheet

- 18.8 For certification of ozone analyzer and transfer standards, open the "Cert Report Ozone". An example of an Ozone Certification Report is shown in Figure 29. Print this report for filing and also provide a copy to the customer.

CALIFORNIA AIR RESOURCES BOARD

OZONE CERTIFICATION REPORT

To: STANDARDS LABORATORY
JERRY FREEMAN

From: JERRY FREEMAN
OPERATIONS PLANNING & ASSESSMENT

Log Number: 2013191

Calibration Date: 7/15/2013
Report Date: 7/17/2013

IDENTIFICATION

Instrument: API 400E, PRIMARY STANDARD

Property No.: 20092613

Serial No.: 2271

Previous Log No.: 2013001

Bar Code No.: 20092613

Elevation: 25.0'

Property of: STANDARDS LABORATORY

Site Name: MLD Standards Lab

Site Number: 34-299

Location: 1927 13th Street
Sacramento, CA
95811

CALIBRATION STANDARD	ID Number
NIST STANDARD REFERENCE PHOTOMETER	4

CALIBRATION RESULTS

COMPONENT	OZONE
Instrument Range	1000 ppb
Air Flow Rate, SLP/SCCM	809
Air Flow Setting	813
Best Fit Linear Regression	Slope: 1.0031 Intercept: (O3 = ppm): 0.00018991
Deviation From True (%)	0.31
Change From Previous Calibration (%)	0.06%
Previous Calibration Date	7/12/2013

Certification Expires: 1/13/2014

Certification Equation:

True Ozone Conc = 0.9982 x (Net Display) + 0.00031 ppm

AIR RESOURCES BOARD

Calibrated by: _____

Checked by: _____

O3_Cal_Data

Cert Report w-gas

Cert Report Ozone

Cert Report Ver

CertSummary

Figure 29 Ozone Certification Report

- 18.9 For verification of ozone primary standards, open the "Cert Report Ver". An example of an Ozone Verification Report is shown in Figure 30. Print this report for filing and also provide a copy to the customer.

CALIFORNIA AIR RESOURCES BOARD
OZONE VERIFICATION REPORT

To: STANDARDS LABORATORY
JERRY FREEMAN

From: Jerry Freeman
Data Analysis and Special Projects

Log Number: **2E+06**

Calibration Date: 7/15/2013
Report Date: 7/17/2013

IDENTIFICATION

Instrument:	API 400E, PRIMARY STANDARD
Property No.:	20092613
Serial No.:	2271
Previous Log No.:	2013001
Bar Code No.:	20092613
Elevation:	25.00'
Property of:	STANDARDS LABORATORY

Site Name:	MLD Standards Lab
Site Number:	34-299
Location:	1927 13th Street Sacramento, CA 95811

CALIBRATION STANDARD	ID Number
NIST STANDARD REFERENCE PHOTOMETER	4

VERIFICATION RESULTS

COMPONENT	OZONE
Instrument Range	1000 ppb
Display	0 - 1
Air Flow Rate, SLPM	809
Air Flow Setting	813
Best Fit Linear Regression	Slope: 1.0031 Intercept: (ppm): 0.0002
Deviation From True (%)	-0.31%
Change From Previous Calibration (%)	0.06%
Previous Calibration Date	7/12/2013

AIR RESOURCES BOARD

Comments: This primary standard meets the criterion set by the EPA for a successful ozone photometer verification. The calculated slope of the comparison data is within 3% of the true and the intercept is less than 3.0 ppb. The resultant slope and intercept values should not be used for correcting any ozone Standard's data or any ozone air monitoring data.

Certification Equation:

True Ozone Conc = **0.9969** x (Net Display) **0.0002** ppm

Verified by: _____

Checked by: _____

O3_Cal_Data
Cert Report w-gas
Cert Report Ozone
Cert Report Ver
CertSummary

Figure 30 Ozone Verification Report

19.0 Routine Service Checks

- 19.1 Weekly or as required. Perform the SRP system adjustments as outlined in Section 7 of “EPA’s, Standard Operating Procedures for Verification of EPA’s Ozone Standard Reference Photometer” on an as-needed basis or following any maintenance. Complete the SRP Operating Characteristics Data Sheet with the data generated by the procedures.
- 19.2 For convenience, an electronic version of the verification procedures pertaining to SRP #4 was created to allow electronic storage of the completed forms. See Figure 31 for the file location of the forms.

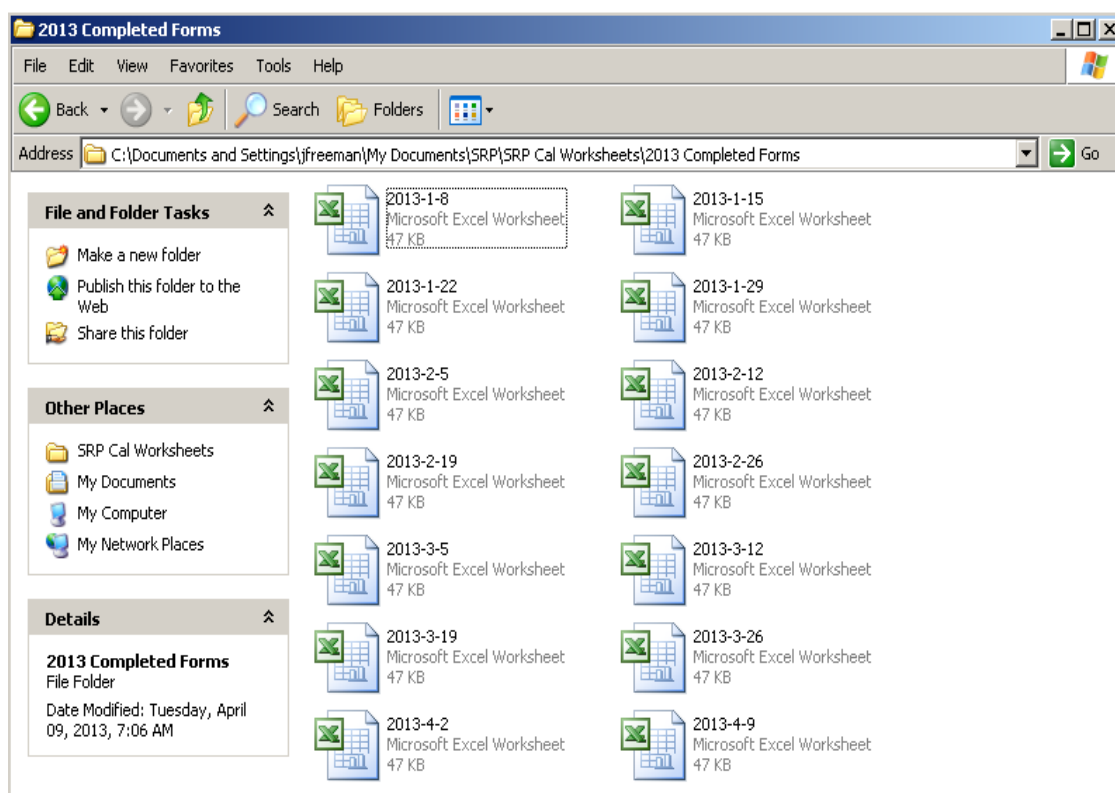


Figure 31 Maintenance Forms

- 19.3 The naming format for the completed forms is “Year-Month-Day”. To start a new form, open the most current form and perform a “Save-As” function using the current day as the file name. Log the appropriate data by following the directions on the form and save when completed.
- 19.4 Quarterly, a verification is performed between SRP #4 and our ozone primary standard, an API 400E. A control chart is derived from historical comparisons. The control chart illustrates possible warning and out of control limits conditions.

- 19.5 For warning limit violations, investigations are initiated into the quality of house air and parameters associated with the SRP.
- 19.6 For out of control limit violations, no instrument comparisons are allowed until the cause is found. Perform all the checks associated with a warning limit violation. If no cause is revealed, a request for a comparison with another SRP should be made to U.S. EPA. No further instrument comparisons should be made until the results of the SRP comparison is analyzed.
- 19.7 Annual practices. A second SRP (SRP # 7) is used to verify the proper operation of SRP # 4. Originally, U.S. EPA was performing this verification annually. However, the SRP's historical accuracy and stability has allowed it to be verified on a biennial basis. U.S. EPA ships SRP # 7 and ARB Standards Laboratory personnel perform the comparison.

20.0 Preventive Maintenance and Repairs

- 20.1 If the SRP fails to pass a diagnostic test or a fault is identified during a routine check, contact U.S. EPA, Air Pollution Prevention and Control Division, Research Triangle Park, North Carolina. This group has managerial control over the SRP Program including spare parts and technical expertise of the equipment.
- 20.2 The guest ozone instruments submitted for certification/verification should be in peak operational condition. However, if an instrument is deficient, contact the owner for resolution.

21.0 Troubleshooting

- 21.1 SRP system adjustments are outlined in chapter 7 of "EPA, Standard Operating Procedures for Verification of EPA's Ozone Standard Reference Photometer", June 29, 2012 Rev 1.04. An extensive troubleshooting guide is included in Table 8.1.
- 21.2 There are several types of ozone analyzers submitted for certification. Brief operating instructions are provided in the Ozone Equipment Reference Binder as shown in Figure 6. For detailed information, the equipment operation/maintenance manuals are usually accessible through the manufacturer's web site. A customer support service may also be available.
- 21.3 The zero air supply is critical for the operation of the SRP. Verify that the proper air pressure and volume is available before starting any test. Contamination could cause significant damage to the SRP and guest instruments under test.

22.0 References

- EPA's, Standard Operating Procedures for Verification of EPA's Ozone Standard Reference Photometer, June 29, 2012 Rev 1.04.
- Ozone Equipment manuals. Many of these manuals may be accessible on the manufacturer's websites.
- Ozone Equipment Reference Binder

23.0 Data Calculations

- 23.1 Prior to use, an ozone transfer standard must be verified by establishing a quantitative calibrations relationship between the transfer standard and the higher level ozone concentrations obtained by the UV calibration procedure as specified in Appendix D of 40 CFR, Part 50 (EPA 1979). The calibrations procedure is extracted from U.S. EPA's Transfer Standards for Calibration of Air Monitoring Analyzers for Ozone, November, 2010. For the Standards Laboratory's purpose, all the calculations are performed using Excel spreadsheets when the data is entered (see Section 18).
- 23.2 The calibrations relationship shall consist of the average of 6 individual comparisons of the transfer standard to the authoritative UV ozone standard system. Each comparison must be carried out on a different day.
- 23.3 Each comparison shall consist of at least 6 comparison points. These points are evenly spaced over the concentration range of the transfer standard, including 0 and 90 ± 5 percent of the upper range limit. Compute the slope and intercept of the 6 comparison points by the least squares linear regression of the transfer standard output (either a generated ozone concentration or a concentration assay) and the UV authoritative ozone standard.

23.4 For the 6 comparisons compute the average slope ($\overline{m}$):

$$\overline{m} = \frac{1}{6} \sum_{i=1}^6 m_i$$

Equation 1

and the average intercept ($\overline{I}$):

$$\overline{I} = \frac{1}{6} \sum_{i=1}^6 I_i$$

Equation 2

where m_i and I_i are the individual slopes and intercepts, respectively, of each comparison regression.

23.5 Compute the relative standard deviation of the 6 slopes, (s_m):

$$s_m = \frac{100}{\overline{m}} \sqrt{\frac{1}{5} \left[\sum_{i=1}^6 (m_i)^2 - \frac{1}{6} \left(\sum_{i=1}^6 m_i \right)^2 \right]} \%$$

Equation 3

and the standard deviation S_I defined in Equation 4 for the 6 intercepts:

$$s_I = \sqrt{\frac{1}{5} \left[\sum_{i=1}^6 (I_i)^2 - \frac{1}{6} \left(\sum_{i=1}^6 I_i \right)^2 \right]}$$

Equation 4

23.6 The value of s_m must be $\leq 3.7\%$, and s_I must be ≤ 1.5 (ppb).

23.7 If the transfer standard meets the specifications of paragraph 23.6, compute the verification relationship as:

$$\text{Std. O}_3 \text{ conc.} = \frac{1}{\overline{m}} (\text{Indicated O}_3 \text{ conc.} - \overline{I})$$

Equation 5